

Deep inelastic scattering on polarized lithium: the physics case for ${}^6\text{Li}$ and ${}^7\text{Li}$ beams at the Electron-Ion Collider

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Abstract. Polarized ${}^6\text{Li}$ and ${}^7\text{Li}$ beams would give the Electron-Ion Collider (EIC) spin-polarized nuclei that are light enough for ab initio computation yet carry genuine nuclear structure: ${}^6\text{Li}$ is a spin-1 nucleus well described as a deuteron bound to an inert α particle, ${}^7\text{Li}$ its spin-3/2 partner, an α plus a triton. We review why lithium is the practical choice — an alkali atom that optical pumping polarizes, a pair of cluster nuclei whose spin sits on the light cluster, and ions whose spin dynamics in the ring are deuteron-like — and why the physics requires both polarization and a collider. Starting from the spin-1 formalism of inclusive deep inelastic scattering we develop the four measurements the programme is built around: the tensor structure function b_1 , the double-helicity-flip structure function Δ (nuclear gluonometry), the polarized EMC effect on the bound proton of ${}^7\text{Li}$, and spectator-tagged scattering on the embedded deuteron. For each we state what the beam must provide and what one EIC year (10 fb^{-1} per nucleon) reaches. Tensor polarization is the scarce resource, the transverse spin direction the natural one, and the spectator-tagged and coherent channels the ones that fix the far-forward requirements.

1 · Introduction

The EIC project scope provides polarized electrons, protons and ${}^3\text{He}$; polarized deuterons are a later addition, and no heavier polarized ion appears in the conceptual design or the Yellow Report [1,2]. The EPIOS white paper makes the case for polarized ${}^6\text{Li}$ and ${}^7\text{Li}$ as "a desirable addition", listed with ${}^2\text{H}$ and ${}^3\text{He}$ as critical technology [3], and the ANL source development specifies the beam: an atomic-beam source with optical pumping at 671 nm feeding EBIS charge breeding, with source targets $P_z \geq 0.90$, $P_{zz} \geq 0.80$ and $\geq 10^{13}$ atoms/s [4]. Four measurements motivate the effort, each addressing a gap in the literature: no b_1 prediction or measurement exists for any nucleus beyond the deuteron; no gluon-transversity projection exists for any EIC target; no study of α -tagged deep inelastic scattering (DIS) on lithium exists; and no lithium-beam luminosity or polarization parameter set has been published.

This note is the physics case behind those measurements, written for colleagues joining the programme. Section 2 argues for lithium as an atom, as a nucleus and as a stored ion; Section 3 for polarization and for a collider; Section 4 develops the spin-1 formalism; Section 5 the four observables; Section 6 collects what each requires of the beam and what the programme's simulations project for one EIC year. The projections themselves — rates, bins, estimators and the reconstructed-level results — are the subject of the companion report [5].

2 · Why lithium

2.1 · The atom: optical pumping and a proven source

Lithium is the lightest alkali metal and its atom is hydrogen-like: a single 2s valence electron whose D lines near 671 nm (670.96 and 670.98 nm in vacuum for ${}^7\text{Li}$) lie within the range of commercial single-frequency diode lasers, and whose ground-state hyperfine splittings — 228.2 MHz for ${}^6\text{Li}$, 803.5 MHz for ${}^7\text{Li}$ [6] — are small enough for one laser with an electro-optic sideband to address both hyperfine levels. Circularly polarized light pumps the atoms into a single magnetic substate, the hyperfine coupling transfers the electron polarization to the nucleus, and radio-frequency transitions between Zeeman sublevels set any combination of P_z and P_{zz} [4]. Boron, the other light candidate with a large nuclear spin, "presents a challenge for optical pumping" [3]. Lithium's low ionization energy (5.39 eV [6]) eases ionization, and fully stripped Li^{3+} is within reach of the EBIS being commissioned to charge-breed the polarized ${}^3\text{He}^{++}$ beam, whose design goal is 2×10^{11} ions per pulse at $\geq 70\%$ polarization [7].

Both isotopes are stable and abundant (7.6% ${}^6\text{Li}$, 92.4% ${}^7\text{Li}$), and they are the only stable isotopes of one element with $J = 1$ and $J = 3/2$: the stable spin-1 nuclei are exactly ${}^2\text{H}$, ${}^6\text{Li}$ and ${}^{14}\text{N}$, and the stable spin-3/2 nuclei below $A = 30$ are ${}^7\text{Li}$, ${}^9\text{Be}$, ${}^{11}\text{B}$, ${}^{21}\text{Ne}$ and ${}^{23}\text{Na}$ [8]. One source, one laser system and one injector therefore serve a spin-1 and a spin-3/2 programme. EPIOS's list of where an exotic-glue discovery is most likely — "spin-1 ${}^6\text{Li}$ or ${}^{14}\text{N}$, or spin-3/2 ${}^7\text{Li}$ or ${}^{23}\text{Na}$ " [3] — reduces, once optical pumping is required, to lithium and sodium, and lithium is the lighter, cluster-structured, ab initio-computable one.

The source physics has a forty-year history. The Heidelberg tandem delivered ${}^6\text{Li}^{3+}$ at vector polarization 0.57–0.65 in 1977 [9]; by 1987 its atomic-beam source reached $P_z, P_{zz} \geq 0.85$ and accelerated ${}^7\text{Li}$ at $P_{zz} = 0.69$ on target [10]; the optically pumped Florida State source delivered more than 150 nA of ${}^6\text{Li}^{3+}$ at $P_{yy} = -1.33 \pm 0.02$ with no polarization loss in the linac [11]; and polarized ${}^6\text{Li}$, ${}^7\text{Li}$ and ${}^{23}\text{Na}$ beams served two decades of nuclear-reaction studies [12]. New for the EIC are scale — $\geq 10^{13}$ atoms/s, two to three orders of magnitude beyond these sources — and transport: preservation of the polarization through charge breeding and acceleration beyond 100 GeV/u [3,4].

2.2 • The nuclei: the spin sits on a light cluster

The two stable lithium isotopes are the lightest nuclei beyond $A = 3$ in which a polarized light cluster is bound to an inert core: in ${}^6\text{Li}$ an α particle ($J = 0, T = 0$) and a deuteron-like cluster bound by 1.47 MeV against $\alpha+d$ breakup, in ${}^7\text{Li}$ an α and a triton bound by 2.47 MeV [8]. Variational Monte Carlo (VMC) with realistic two- and three-nucleon forces makes this quantitative: the α -d overlap of the ${}^6\text{Li}$ ground state integrates to $N_{\alpha d} = 0.86$ (0.846 S-wave, 0.017 D-wave) and the α -t overlap of ${}^7\text{Li}$ to $N_{\alpha t} = 1.00$ [13]; the no-core shell model with continuum gives an asymptotic D/S ratio of -0.027 for the $d+\alpha$ configuration [14]. In a fully polarized ${}^6\text{Li}$ the α contributes nothing and the two valence nucleons carry the spin: six-body VMC gives valence numbers 1.924 up and 1.076 down, a polarization of 0.85 [13,15]; three-body $\alpha+n+p$ models, in which $\alpha+d$ accounts for 60–70% of the wave function, give above 90% [16]; and E155, which used ${}^6\text{LiD}$ as its deuteron target, adopted 87% for the effective deuteron polarization [17]. ${}^6\text{Li}$ is therefore a polarized deuteron in the field of an α particle, whose tensor observables can be set against the free deuteron's to isolate the medium effect. The only other stable spin-1 ground state, ${}^{14}\text{N}$, shares its spin among p-shell nucleons and lies beyond the reach of exact quantum Monte Carlo ($A \leq 12$) [18].

In ${}^7\text{Li}$ the spin is carried by the unpaired proton of the triton cluster, the neutrons being paired [19]. VMC gives proton numbers 1.934 up and 1.066 down and neutron numbers 1.981 and 2.019 [13], i.e. $P_p = 0.87$ and $P_n = -0.04$ (0.866 and -0.037 as quoted in the JLab proposal [20]); the shell model already gives $P_p = 13/15$. Polarized ${}^7\text{Li}$ is thus to the proton what polarized ${}^3\text{He}$ is to the neutron, with the property, decisive for the polarized EMC effect, that the proton sits in a nucleus of normal density. Cloët, Bentz and Thomas set the selection criteria — $|P_n| \ll |P_p|$ and light, because g_1^A is carried by the valence nucleons and is suppressed relative to F_2^A by roughly $1/A$ — and list ${}^7\text{Li}$ among the candidates "where the nuclear polarization is largely dominated by the valence proton" [21]; the gluon-spin study names ${}^7\text{Li}$ "the most promising case for experimental investigation" [22].

Two further properties matter. The cluster structure makes the nuclei computable: Green's function Monte Carlo with AV18+IL7 reproduces both binding energies (-31.93 vs -31.99 MeV for ${}^6\text{Li}$, -39.15 vs -39.25 MeV for ${}^7\text{Li}$), the charge radii to within 3% and, with two-body currents, the magnetic moments (0.835 vs $0.822 \mu_N$; 3.24 vs $3.256 \mu_N$) [18,19], so the nuclear corrections entering every extraction — effective polarizations, momentum distributions, spectral functions — carry controlled uncertainties rather than model spread. The quadrupole moment of ${}^6\text{Li}$ is the exception: the measured $Q = -0.0806 \text{ fm}^2$, against $+0.286 \text{ fm}^2$ for the free deuteron [8], arises from a delicate cancellation between the deuteron's own moment and the D-wave part of the α -d relative motion [15,23], and theory reproduces only its smallness. That smallness is an asset — ${}^6\text{Li}$ is nearly spherical, which bounds the deformation term of the coherent channel (§5.2) — while ${}^7\text{Li}$, whose p-shell triton orbit gives $Q = -4.00 \text{ fm}^2$ [8], is the deformed counterpart. The $\alpha+d$ versus $\alpha+t$ comparison changes only the cluster.

2.3 • The ion: deuteron-like spin dynamics and deuteron kinematics

In the ring, ${}^6\text{Li}$ is the easy case. Its gyromagnetic anomaly, $G \approx -0.18$, is as small as the deuteron's (-0.14), so it crosses only 81 depolarizing resonances on the way to top energy, against 63 for the deuteron and 2457 for ${}^3\text{He}$ [24]; like the deuteron it is handled with a partial solenoid snake and tune jumps, full Siberian snakes being impractically long for so small a G [3,24], so the deuteron techniques [25] carry over. ${}^7\text{Li}$ ($G \approx +1.52$) crosses 573 resonances and needs partial snakes with jump quadrupoles [3,24].

The energies follow from the collider, not from the ion. The hadron and electron rings must share a revolution period, so the hadron Lorentz factor is fixed by the circumference and the magnets supply whatever rigidity that γ demands, up to the 917 T·m cap: the ion menu is γ -matched to the proton's, an isolated point at ≈ 41 GeV/u and a continuous band from ≈ 99 GeV/u up to the species' rigidity limit of $275 \text{ GeV} \times Z/A$ [3,26]. For ${}^6\text{Li}$ that is 40.8 GeV/u and 99.5–137.5 GeV/u, for ${}^7\text{Li}$ 40.8 and 99.5–117.9 GeV/u. Because ${}^6\text{Li}$ has $A/Z = 2$ it runs at the same per-nucleon energy and magnet settings as the deuteron, so $e+d$ and $e+{}^6\text{Li}$ can be compared at identical kinematics. The stable spin direction in the arcs is vertical, so transverse polarization is native at the interaction point while longitudinal polarization needs spin rotators whose lithium settings are undefined [1,3].

3 • Why polarization, and why a collider

3.1 • Polarization is the observable

An unpolarized lithium beam would be one more light ion: its inclusive cross section is carried by F_1 and F_2 alone — for an unpolarized target the spin-1 cross section has exactly the spin- $\frac{1}{2}$ form [27] — and it would add an $A = 6$, 7 point to the EMC and shadowing curves that ${}^4\text{He}$ and ${}^{12}\text{C}$ already map. That point is not without interest: the high- x EMC ratios of ${}^6\text{Li}$ and ${}^7\text{Li}$ are in the target list of JLab E12-10-008 [28], motivated by the finding that the ${}^9\text{Be}$ EMC effect follows the local density of its α clusters [29], and the α -tagged structure function of the triton or deuteron cluster versus spectator momentum would measure the α - t and α - d momentum distributions VMC predicts [13]. But every flagship item of the programme is a spin observable, and the two tensor ones need the tensor moment specifically: b_1 is *defined* as the difference between the $m = 0$ and $m = \pm 1$ quark distributions, so averaging over m erases it, and Δ needs both a target spin of at least 1 and a transverse alignment axis to define an azimuth (§4). The polarized EMC effect is a statement about g_1 , and tagged DIS reads the spin of the cluster (§5). Polarized lithium would be the first polarized ion at a collider with a nuclear core — the first place where the spin structure of a *nucleus*, rather than of a nucleon, is studied at collider kinematics — and tensor polarization, on which both b_1 and Δ depend, is the scarcer resource.

3.2 • Fixed target or collider

Polarized lithium has existed for decades as a fixed target, but only in the solid state: dynamically polarized ${}^6\text{LiD}$ was the deuteron target of SLAC E155 and of COMPASS, ${}^7\text{LiH}$ has been polarized at Saclay (47% for ${}^7\text{Li}$, 56% for the protons) [30], and ${}^7\text{LiD}$ is the material of the approved CLAS12 polarized-EMC experiment [17,20,31]. In these compounds the lithium polarizes along with the hydrogen isotope because all spin species share one spin temperature: COMPASS reached deuteron polarizations of +54%/−47%, whence $51 \pm 3\%$ for ${}^6\text{Li}$ and $92 \pm 4\%$ for the ${}^7\text{Li}$ admixture [31]; E155 measured ${}^6\text{Li}$ at 97% of the deuteron's [17]; in ${}^7\text{LiD}$ at 5 T the CLAS12 proposal expects 65–80% for ${}^7\text{Li}$ [20]. Three limits are structural. (i) *Dilution*: only $f \approx 0.35$ – 0.45 of the nucleons in ${}^6\text{LiD}$ are polarizable once the helium coolant is counted — better than 0.22 for ND_3 and 0.15 for NH_3 , far from the 1.0 of a stored beam [17,31]. (ii) *Tensor polarization*: in thermal equilibrium it follows from the vector one, $P_{zz} = 2 - \sqrt{4 - 3P_z^2}$, so $P_z = 0.5$ gives $P_{zz} = 0.20$; radio-frequency "hole burning" reaches 29–37% in deuterated butanol at the cost of the vector polarization, and the JLab b_1 experiment is conditionally approved on demonstrating 30% [32]. An atomic-beam source prepares the tensor moment directly: HERMES ran its gas target at $P_{zz} = +0.89$ and -1.65 , "a combination which is not possible for solid-state polarized targets" [33]. (iii) *Spectator loss*: a spectator α of 100–300 MeV/c carries only 1.3–12 MeV in the target rest frame and stops within 10–250 μm of lithium deuteride [34], inside cells 2–5 cm long at JLab and 30–60 cm at COMPASS, and a coherent ${}^6\text{Li}$ recoil at $|t| = 0.05 \text{ GeV}^2$ has only 4.5 MeV. The detectors that do tag slow spectators, BoNuS (protons from 70 MeV/c) and

ALERT (α from 230 MeV/c), work with unpolarized gas at a few atmospheres [35,36]; polarized lithium outside dynamically polarized solids has never been used in lepton scattering.

The collider removes each limit at once. The beam is pure lithium, its polarization prepared atom by atom and measured in the ring; the spectator α or triton carries the beam momentum per nucleon into the far-forward spectrometers at a few milliradians; the intact coherent recoil can be tagged down to $p_T \approx 0$ at the secondary focus of a second interaction region [37]; and $\sqrt{s_{NN}}$ up to 99 GeV for $e(18 \text{ GeV}) \times {}^6\text{Li}(137.5 \text{ GeV/u})$ opens x down to $\sim 10^{-4}$ at $Q^2 \geq 1 \text{ GeV}^2$, where JLab at 11 GeV stops at $x \approx 0.06$ and HERMES stopped at 0.01 [20,33] — the region where gluons dominate, where HERMES saw its large b_1 , and where coherence holds. What stays with the fixed target is luminosity: $10^{35} \text{ cm}^{-2} \text{ s}^{-1}$ (CLAS12 at 10–15 nA, Hall C at 115 nA) [20,32] against an electron–nucleon luminosity of $10^{33}\text{--}10^{34} \text{ cm}^{-2} \text{ s}^{-1}$ at the EIC [1,2]. The high- x polarized EMC effect and the high- x b_1 of the deuteron are therefore rightly JLab measurements; the EIC takes the low- x and high- Q^2 regions, the tensor observables at full polarization, and everything that needs a detected spectator or recoil (Table 1).

MEASUREMENT	AT A FIXED TARGET	WHAT ONLY THE COLLIDER ADDS
Polarized EMC effect, ${}^7\text{Li}$ (g_1^A)	Approved: JLab E12-14-001 (CLAS12, polarized ${}^7\text{LiD}$, $0.06 < x < 0.8$, $1 < Q^2 < 15 \text{ GeV}^2$, $P_z \approx 0.8$ assumed) [20]	$x < 0.06$; the Q^2 evolution that carries the gluon-spin EMC effect [22]; an undiluted beam; α -tagging of the struck triton and its virtuality
Tensor b_1 , ${}^6\text{Li}$	Deuteron only: E12-13-011 (ND_3 , $0.16 < x < 0.49$, $P_{zz} \geq 0.3$ required) [32]; ${}^6\text{LiD}$ would give $P_{zz} \approx 0.2$ with the deuteron term to be subtracted, and no such measurement is proposed	x down to 10^{-3} , where HERMES found $b_1 \approx 0.1$; pure-tensor fills at $P_{zz} \approx 0.6\text{--}0.8$; spin axis along $\mathbf{q}$ [38]
Exotic glue Δ , ${}^6\text{Li}$ (${}^7\text{Li}$)	Letter of intent only: LOI12-16-006 (${}^{14}\text{N}$ in NH_3 , $x < 0.3$, 2–12% alignment, in-plane spectrometers — $\cos 2\phi$ "difficult with a fixed target") [39]	Transverse tensor polarization ≥ 0.6 with full 2π azimuthal acceptance; the gluon-dominated region $x \lesssim 0.1$ and a Q^2 lever arm
Spectator-tagged DIS (α , t)	Not feasible: spectators stop within 0.25 mm of a solid target; no polarized lithium gas target exists	Far-forward detection of α and t at beam velocity; pole extrapolation to the free cluster [40,41]
Coherent intact- ${}^6\text{Li}$ channel	Not feasible: needs $x \lesssim 0.01$ and a 2–9 MeV recoil	A near-beam tag of the intact recoil: a dedicated tagging optics at IP6 or the IR-8 secondary focus [37]

Table 1 — what a fixed target covers and what only the collider adds, measurement by measurement.

4 • Formalism

4.1 • Deep inelastic scattering

An electron with four-momentum k scatters from a nucleus of momentum P by exchanging a virtual photon $q = k - k'$. The kinematic invariants, in the per-nucleon convention used throughout the programme (energies and structure functions per nucleon, $x \leq 1$), are

$$Q^2 = -q^2, \quad x = \frac{Q^2}{2P \cdot q}, \quad y = \frac{P \cdot q}{P \cdot k}, \quad Q^2 = x y s, \quad W^2 = M^2 + Q^2 \frac{1-x}{x}$$

where x is the momentum fraction of the struck parton, y the energy-transfer fraction and W the invariant mass of the hadronic final state. The unpolarized cross section is carried by two structure functions,

$$\frac{d^2\sigma}{dx dQ^2} = \frac{4\pi\alpha^2}{x Q^4} \left[\left(1 - y + \frac{y^2}{2}\right) F_2(x, Q^2) - \frac{y^2}{2} F_L(x, Q^2) \right], \quad F_1 = \frac{F_2}{2x(1+R)}$$

with $R = \sigma_L/\sigma_T$, for which the programme's projections use a placeholder of the right magnitude (0.14–0.18 at the kinematics of interest) and carry the SLAC/E143 R1998 world fit [42] behind a switch whose effect is quantified in §5.2. In the parton model $F_2 = \sum_q e_q^2 x[q(x) + \bar{q}(x)]$: the cross section measures quark distributions weighted by squared charge.

For a nucleus we use F_2^A/A , so rates are quoted against luminosity per nucleon. Polarization introduces additional structure functions whose number and character depend on the target spin.

4.2 • Spin-1 targets

A spin- $1/2$ target contributes two spin structure functions, g_1 and g_2 . The principal observable is g_1 , the helicity-weighted quark distribution, measured through the double-spin asymmetry $A_{\parallel} \approx D \cdot g_1 / F_1$ (D is a kinematic depolarization factor) with both beams longitudinally polarized and the electron helicity reversed bunch by bunch.

A spin-1 target carries additional structure. Its spin state along a chosen axis is specified by three populations p_+ , p_0 , p_- of the projections $m = +1, 0, -1$, and two moments characterize the fill:

$$P_z = p_+ - p_- \in [-1, 1], \quad P_{zz} = p_+ + p_- - 2p_0 = 1 - 3p_0 \in [-2, 1]$$

P_z is the familiar vector polarization. P_{zz} , the *tensor* or alignment polarization, is specific to spin ≥ 1 : it distinguishes spin along $\pm z$ from spin in the transverse plane ($m = 0$) and is nonzero even when equal numbers point up and down. An unpolarized beam has $(1/3, 1/3, 1/3)$ and both moments vanish; a pure $m = 0$ beam has $P_z = 0$ but $P_{zz} = -2$ (Figure 1).

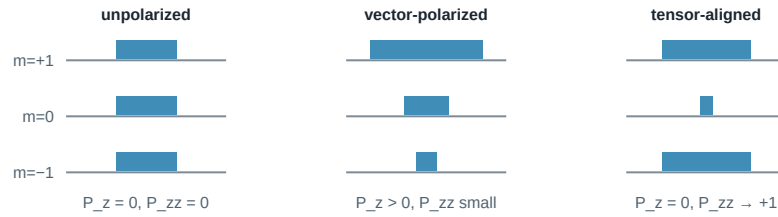


Figure 1 — populations of the three spin projections for an unpolarized, a vector-polarized and a tensor-aligned spin-1 fill. The tensor moment measures the $m = \pm 1$ population against $m = 0$ and is independent of the vector one.

Hoodbhoy, Jaffe and Manohar showed [27] that a spin-1 target has, beyond F_1 , F_2 , g_1 , g_2 , four tensor-polarized structure functions b_1 – b_4 , and Jaffe and Manohar identified one more, the double-helicity-flip function Δ [43]. At leading twist the tensor sector reduces to b_1 (with $b_2 = 2x b_1$ as its Callan–Gross partner) and Δ . For unpolarized electrons on a target whose spin axis makes angle θ_m with the beam, the cross section for each pure m state is [27,43]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left(F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_m \Delta(x, Q^2) \cos 2\phi \right]$$

$$a_m = \frac{1}{4} c_m (3 \cos^2 \theta_m - 1), \quad c_m = 3|\lambda_m| - 2 \rightarrow (1, -2, 1) \text{ for } m = (+1, 0, -1)$$

where ϕ is the azimuth of the lepton plane about the photon direction, measured from the transverse projection of the spin axis. The b_1 , b_2 terms are written in the programme's transcription; in the Hoodbhoy–Jaffe–Manohar convention as applied by HERMES and by Cosyn et al. [38] ($A_{zz} = -(2/3) b_1 / F_1$ for the axis along the photon) they carry the opposite sign. Only the sign of A_{zz} relative to b_1 is affected; magnitudes, errors and the Δ term are not.

The weights $(1, -2, 1)$ organize all tensor observables: the $m = \pm 1$ states enter together and against $m = 0$, precisely the combination measured by P_{zz} , and averaging over a fill replaces every c_m by P_{zz} . Two handles follow. With the spin axis *longitudinal*, the rate difference between fills gives $A_{zz} \propto b_1$; with it *transverse* ($\theta_m = 90^\circ$), the ϕ distribution of the scattered electron acquires a $\cos 2\phi$ modulation proportional to Δ . Both require tensor polarization; neither requires polarized electrons.

^7Li , with spin $3/2$, has four populations and three moments (vector, tensor/quadrupole, octupole). Its rank-0 and rank-1 sectors are exact analogues of the spin- $1/2$ case and host the polarized EMC measurement; its rank-2 sector is the formal analogue of the spin-1 tensor sector, for which no structure-function basis has been published.

5 • The observables

5.1 • The tensor structure function b_1

In parton language b_1 compares quark distributions in an aligned versus a transverse nucleus [27],

$$b_1(x) = \frac{1}{2} \sum_q e_q^2 \left[q^{(0)}(x) - \frac{q^{(+1)}(x) + q^{(-1)}(x)}{2} \right], \quad \int_0^1 b_1(x) dx = 0 \quad (\text{Close – Kumano, no tensor sea})$$

where $q^{(m)}$ is the quark distribution in a target with spin projection m . For free spin- $\frac{1}{2}$ nucleons, or a pure S-wave bound state, all m states are equivalent and $b_1 = 0$, so a nonzero b_1 measures directly how binding modifies the partonic content: D-wave admixture, pion exchange, hidden-colour components, or a tensor-polarized sea, which would violate the Close–Kumano sum rule [44].

The only existing measurement is HERMES on the deuteron [33]: A_{zz} up to 2×10^{-2} over $0.01 < x < 0.45$ and $b_1(x = 0.012) = 0.112 \pm 0.055 \pm 0.028$, unexpectedly large at low x , with a zero crossing near $x \approx 0.2$ and a sum-rule integral 2σ from zero. Convolution calculations predict $|b_1| < 10^{-3}$ at $x \gtrsim 0.2$, an order of magnitude below the data [45], whereas a model with pions and a hidden-colour probability of only 0.15% reproduces it [46]. JLab E12-13-011 will remeasure $0.16 < x < 0.49$ with a solid ND_3 target [32]; the EIC adds the low- x region and the Q^2 lever arm. The beam is delivered in fills of different tensor polarization ("tensor thirds"); with the estimator $(N_+ + N_- - 2N_0)$ over matched luminosities all m -independent physics cancels (§6). At collider kinematics a clean extraction favours orienting the spin axis along the momentum transfer $\mathbf{q}$ rather than along the beam [38].

5.2 · $\Delta(x, Q^2)$: exotic glue

The $\cos 2\phi$ term corresponds to the virtual photon flipping its helicity by two units between absorption and emission in the forward Compton amplitude. A spin- $\frac{1}{2}$ particle cannot absorb two units of helicity, so neither quarks nor bound nucleons generate Δ at leading twist; gluons can. Δ is what the gluon transversity distribution — an interference of gluon helicity states that exists only for $J \geq 1$ — generates at order α_s through a photon–gluon convolution [43,47], so a nonzero Δ in a nucleus is direct evidence for glue not inherited from individual nucleons, hence "exotic glue". It has never been measured; the one prior experimental document is a JLab letter of intent [39].

The measurement uses unpolarized electrons on a tensor-polarized nucleus whose alignment axis lies in the transverse plane, and fits the azimuthal distribution about that axis. Averaged over the fill, the amplitude and its inversion back to Δ are

$$A^{\cos 2\phi} = -\frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2$$

Writing $F_2 = 2x(1+R)F_1$ removes both structure functions from the ratio Δ/F_1 except through R , which is the one structure-function input that reaches the physics result — F_2 cancels exactly. At the four kinematic bins of the companion projection ($x = 0.011\text{--}0.14$, $Q^2 = 1.1\text{--}14 \text{ GeV}^2$), taking R from the R1998 fit rather than a simplified form moves Δ/F_1 by +18, +19, +8 and −4%, and at the top beam configuration it reduces the luminosity the fast simulation needs for 5σ on $\Delta/F_1 = 10^{-3}$ from 275 to 155 fb $^{-1}$ /u [5].

The expected size spans per mille to per cent, and the only absolute normalizations available are moment estimates: a bag-model calculation for the spin- $3/2$ Δ^{++} baryon, $\int x \Delta dx = -0.012 \alpha_s$ [47], and lattice-QCD calculations of gluon transversity in the ϕ meson (clearly nonzero, at an unphysical pion mass) [48], with the analogous matrix element roughly ten times suppressed in the deuteron [49]. Percent-level asymmetries appear only in proton–deuteron Drell–Yan under the maximal assumption $\Delta_T g = \Delta g$, which its authors call optimistic [50]. No calculation exists for ${}^6\text{Li}$ or any $A > 2$ nucleus, and the literature brackets the nuclear Δ both ways: suppressed, as in the lattice deuteron, or enhanced by binding in analogy with the EMC effect [3]. The companion report transfers the bag moment to ${}^6\text{Li}$ as a labelled scenario, and finds that one EIC year measures the amplitude at $\geq 20\sigma$ per bin [5].

The coherent channel. The same modulation can be measured in coherent diffraction, $e {}^6\text{Li} \rightarrow e' X {}^6\text{Li}(\text{g.s.})$, where the nucleus stays in its ground state and recoils intact. There the modulation mixes two mechanisms: the nuclear *shape*, a deformed gluon density modulating the diffractive slope (computed for polarized deuterons in the only existing calculation of this kind [51]), and exotic glue proper, which persists at small x , where it connects to the elliptic-gluon programme [52]. ${}^6\text{Li}$ discriminates between them because its quadrupole moment is anomalously small (§2.2): the shape term is bounded and linear in t , so a flat-in- t modulation cannot be a shape effect. Detecting the intact recoil is the difficulty. The nucleus keeps the beam's rigidity and emerges within a milliradian or two of the beam, so the only handle is its angle against the

10σ beam envelope and the Roman-Pot aperture, and the tagged fraction is an exponential in the square of the beam divergence, $\exp[-B(10\sigma_0 \cdot A \cdot p_u)^2]$. With the Yellow Report's divergences the fraction is 5×10^{-7} , 8×10^{-26} and 4×10^{-13} at the three ${}^6\text{Li}$ configurations, and the measured pot aperture is no wider [5,26]: the channel does not exist at IP6 with any published optics. It requires either a dedicated tagging optics — β^* raised so that the envelope sits at $p_T = 1/\sqrt{B} \approx 0.14$ GeV, where the tagged fraction is $1/e$, with pots that follow — or the secondary focus of the second interaction region, which reaches $p_T \approx 0$ [37]. The channel is developed, and its physics case made, in the companion report [5].

5.3 • The polarized EMC effect: g_1 of a bound proton

The EMC effect — the ~ 10 – 20% suppression of F_2 per nucleon in a nucleus at $0.3 \lesssim x \lesssim 0.7$ — has been measured for forty years without a settled explanation. Its spin analogue has never been measured: does the *spin* structure of a bound nucleon change more, less, or the same as its momentum structure? The observable is the polarized EMC ratio

$$\Delta R_A(x) = \frac{g_1^A(x)}{P_p g_1^p(x) + P_n g_1^n(x)}$$

where P_p , P_n are the effective nucleon polarizations from nuclear structure. With $P_p = +0.87$ against $P_n = -0.04$ (§2.2), ${}^7\text{Li}$ is to a good approximation one polarized proton plus six spectators, with computable corrections. Two classes of model differ sharply: Cloët, Bentz and Thomas (mean field, NJL) obtain a polarized EMC effect about twice the unpolarized one [21], Tronchin, Matevosyan and Thomas equal effects [53]. A measurement at the 5% level discriminates between them, and the two map onto the competing mean-field and short-range-correlation pictures of the EMC effect itself. Unique to the EIC is the Q^2 evolution of g_1 , driven by the polarized gluon distribution; the predicted *gluon-spin* EMC effect is larger still, with ${}^7\text{Li}$ the most promising case [22], and the collider measures it free of target-fragment absorption. Both beams are polarized longitudinally and the electron helicity flips bunch by bunch; the error scales as $1/(P_e P_z \sqrt{N})$, so the measurement uses the vector polarization the source delivers most readily. ${}^6\text{Li}$ and the deuteron supply the isoscalar companions, so the three together separate the u- and d-quark spin modification — an inference of this programme, not yet in the literature.

5.4 • Spectator-tagged DIS

Detecting the α spectator, which carries beam velocity into the far-forward spectrometers, identifies the struck object as the deuteron cluster of ${}^6\text{Li}$ or the triton cluster of ${}^7\text{Li}$, so the beam acts as a source of polarized light clusters at controlled virtuality. The framework is spectator tagging [40]: the spectator momentum is measured and extrapolated in its transverse component to the free-cluster pole, recovering free-particle structure with binding effects removed by construction rather than modelled. Tagged *tensor* asymmetries reach $O(1)$ at spectator momenta of ~ 300 MeV/c, where the D-wave dominates [54], far larger than the inclusive asymmetries. The benchmark is the $e^+{}^3\text{He}$ double-tagging study of Frišćić et al. [55], where tagging improved on effective-polarization unfolding by a factor 2–3, and 10 at low x .

Where each fragment is detected follows from magnetic rigidity alone: a beam-velocity fragment bends in the hadron optics according to the ratio of mass-to-charge ratios $R = (m/Z)_{\text{frag}}/(m/Z)_{\text{beam}}$ (Table 2). The (A/Z) ratio approximates it to 5×10^{-3} ; the α is more bound per nucleon than the lithium it leaves, so it sits just below beam rigidity rather than on it.

BEAM	TAG FRAGMENT	R	DESTINATION AT IP6
${}^7\text{Li}$ ($A/Z = 7/3$)	α	0.85571	Roman-Pot main window ($R \in 0.60\text{--}0.95$) — works now
${}^7\text{Li}$	t	1.28971	over-rigid, no coverage
${}^6\text{Li}$ ($A/Z = 2$)	α / d	0.99813 / 1.00452	within 5×10^{-3} of beam rigidity: inside the beam envelope except the p_T tail
${}^6\text{Li}$	p / n	0.50251 / –	Off-Momentum Detectors / ZDC (veto handles)

Table 2 — far-forward routing of the tag fragments at IP6, from the rigidity ratio R alone; the windows are those of Jentsch, Tu and Weiss [41].

In the cluster model of the programme's fast simulation, the ${}^7\text{Li}$ α tag reaches the Roman Pots with $\approx 97\%$ acceptance (95–99% over the wave-function tail scale $\beta = 0.2\text{--}0.4$ GeV and both optics), whereas the ${}^6\text{Li}$ α at 137.5 GeV/u is tagged in

1.9% (high-acceptance optics) or 1.5% (high-divergence) of events, entirely through the p_T tail above the near-beam cut, and the triton of ${}^7\text{Li}$ in 0.6%. Because the ${}^6\text{Li}$ tag is a p_T -tail measurement, its model uncertainty is one-sided upward: the cluster model undershoots the official BeAGLE e+d sample by 2.5 $\times$, 6.9 $\times$ and 28 $\times$ at $p_T > 0.2, 0.3$ and 0.45 GeV. The tagging programme therefore begins with ${}^7\text{Li}$ at IP6, while ${}^6\text{Li}$ α -tagging — and the coherent channel, which faces the same rigidity problem — motivates the near-beam tagging capability of §5.2.

6 • Requirements and reach

All projections use luminosity per nucleon: 10 fb⁻¹/u — 30 weeks at 10³³ cm⁻² s⁻¹ with 60% efficiency in the Yellow Report's accounting [2], "one EIC year" — and 100 fb⁻¹/u for the ten-year programme, at placeholder in-ring polarizations $P_z = 0.7$, $P_{zz} = 0.6$ and $P_e = 0.7$; transport of the source polarization to the ring is an open accelerator question, and every statistical error scales as 1/(P $\sqrt{L}$), so halving P_{zz} costs a factor four in running time. The observables demand different spin configurations and share the luminosity, so each reach assumes the full 10 fb⁻¹/u in its own configuration. The statistical uncertainties behind every projection are

$$\delta A_{\parallel} = \frac{1}{P_e P_z \sqrt{N}}, \quad \delta A_{zz} = \frac{\sqrt{2/N}}{P_{zz}}, \quad \delta A^{\cos 2\phi} = \frac{\sqrt{2/N}}{P_{zz}} \text{ (single state)}, \quad \delta \hat{A} = \frac{2}{P_+ - P_0} \sqrt{\frac{2}{N}} \text{ (two states)}$$

The factor $\sqrt{2}$ in the tensor uncertainties follows from $\langle \cos^2 2\phi \rangle = \frac{1}{2}$ (equivalently, from the three-fill estimator variance); the polarization sits in the denominator because the fill dilutes the physical asymmetry. These formulas, and their behaviour under realistic ϕ -acceptance holes (where a binned fit replaces the naive moment), are validated end to end by the programme's event-generator pseudo-experiments [5]. Tensor observables have a structural advantage when the $m = \pm 1$ -rich and $m = 0$ -rich states are measured with the same detector and compared bin by bin: in their spin-state ratio the ϕ acceptance cancels and relative-luminosity errors enter only as a constant, whereas a single-state fit also measures the detector's own $\cos 2\phi$ acceptance harmonic divided by P_{zz} ; and the two-state pattern $P_+ = +0.6$, $P_0 = -1.2$ is 1.5 $\times$ more precise than a single state at the same source purity. The cancellation requires bunch-by-bunch alternation: a 10⁻³ difference of the acceptance harmonic between the states fakes half a 10⁻³ signal. The polarimetry scale and the tensor radiative corrections do not cancel. Table 3 collects, observable by observable, what the beam must provide and what the simulations project.

OBSERVABLE	BEAM POLARIZATION	ELECTRONS	REACH AT 10 FB ⁻¹ /U	BINDING REQUIREMENT
g_1^A , polarized EMC (${}^7\text{Li}$; ${}^6\text{Li}$ for the isoscalar pair)	Vector $P_z \geq 0.7$, longitudinal (spin rotators). A spin-3/2 spin-temperature fill at $P_z = 0.8$ also carries $P_{zz} \approx 0.55$ and $P_{zzz} \approx 0.11$; the octupole term enters the asymmetry numerator and is corrected through the fill model [20]	Polarized, $P_e \approx 0.7$, helicity flipped bunch by bunch	$\delta \Delta R \approx 3\text{--}4\%$ per x-bin at $x = 0.3\text{--}0.5$ (three energies combined); the CBT-TMT difference is $\approx 1\sigma$ per bin, $\approx 3\sigma$ per bin at $x \approx 0.5\text{--}0.7$ with 100 fb ⁻¹ /u	Polarimetry scale $\delta P_z/P_z \leq 3\%$; P_p from VMC [13]; a Q^2 lever arm at fixed x (gluon-spin EMC)
b_1 via A_{zz} (${}^6\text{Li}$)	Tensor P_{zz} , longitudinal or along q [38]; fills of different P_{zz} ("tensor thirds"), i.e. $m = 0$ bunches from the source	Unpolarized (helicity-averaged)	$\delta A_{zz} \approx 0.9 \times 10^{-4}$ per x-bin (combined over Q^2 and the three energies) at $x < 0.05$ (0.7×10^{-4} at $P_{zz} = 0.8$), rising to $\approx 10^{-3}$ at $x \approx 0.6$, against a scenario $ A_{zz} \approx 10^{-2}$ at $x \leq 0.03$ (HERMES-like b_1 with the 0.29 dilution) or ten times smaller (convolution): both $\geq 10\sigma$ per bin at low x , while $x > 0.3$ is JLab's domain	Relative luminosity between fills at the 10 ⁻⁴ level (a 10 ⁻⁴ offset biases A_{zz} by 1×10^{-4} ; the luminosity-corrected estimator removes it exactly); tensor polarimetry, for which no method is established [3]

Δ via $\cos 2\phi$ (${}^6\text{Li}$; ${}^7\text{Li}$ eligible)	Tensor P_{zz} with the alignment axis transverse (the native vertical direction); fills of opposite P_{zz} sign flip the modulation — the decisive systematic check	Unpolarized	$\delta A = (1.4\text{--}4.5)\times 10^{-4}$ per sweet-spot bin ($x = 0.011\text{--}0.14$, $Q^2 = 1.1\text{--}14 \text{ GeV}^2$) against moment-constrained amplitudes of $(0.44\text{--}0.95)\times 10^{-2}$, i.e. $21\text{--}44\sigma$; a 3×10^{-3} flat amplitude stays $\geq 3\sigma$ per bin even at $P_{zz} = 0.3$. Reconstructed, with the two-state ratio and a hadronic-method y : $\delta A = (0.9\text{--}3.0)\times 10^{-4}$; with a PYTHIA 8 final state through the hadron-side response the bin purities are $0.43\text{--}0.69$ and the reconstructed amplitude $5\text{--}23\%$ below the true-bin value	Spin-state-sorted ratio of $m = \pm 1$ -rich and $m = 0$ -rich bunches alternating bunch by bunch; a hadronic-method y at the $y \approx 0.01\text{--}0.03$ sweet spots, where the electron method alone gives $\delta y/y = 0.5\text{--}1.2$ and the Σ method with a PYTHIA 8 final state and 50 MeV of calorimeter noise $0.32 / 0.22 / 0.29 / 0.15$ (mid configuration; $0.11\text{--}0.24$ for the $x \approx 0.14$ bins at the low one), the noise floor deciding the answer ($0.2 \rightarrow 1.0$ at $y = 0.005$ for $0 \rightarrow 100 \text{ MeV}$, on a hadronic $E - p_z$ sum of $0.2\text{--}0.5 \text{ GeV}$ that no published ePIC study covers); $\delta P_{zz}/P_{zz} \approx 3\%$ scale; tensor radiative corrections not yet calculated
Spectator-tagged DIS	${}^7\text{Li}$ α -tag: any polarization for tagged F_2 ; vector P_z with polarized electrons for tagged g_1 of the bound proton. ${}^6\text{Li}$ α -tag: tensor P_{zz} for the order-unity tagged tensor asymmetries at $p_s \approx 300 \text{ MeV}/c$ [54]	As for the inclusive observable	${}^7\text{Li}$ $\alpha \rightarrow$ Roman Pots ($R = 0.856$), $\approx 97\%$, optics-robust; ${}^6\text{Li}$ α beam-blind at IP6 ($1.9\% / 1.5\%$ at $137.5 \text{ GeV}/u$) $\rightarrow$ near-beam tagging optics or IR-8	Spectator p_T resolution $\sim 10\%$ at $100 \text{ MeV}/c$ [41]; charge identification of α against $A/Z = 2$ fragments in the Roman Pots; pole extrapolation in p_T [40]; the one-sided p_T -tail model uncertainty
Coherent intact- ${}^6\text{Li}$ $\cos 2\phi$	Tensor $P_{zz} \geq 0.6$, transverse	Unpolarized	$x \leq 0.01$, $ t < 0.2 \text{ GeV}^2$. For a tag with a 0.20 GeV near-beam envelope (13.5% of the coherent yield): 1.7×10^7 tagged events, best-bin $\delta A = 1.8\times 10^{-3}$ (5σ at a 0.9% modulation), against a deformation-anchored (a_2) = 0.036 [$0.018\text{--}0.059$] and a flat gluonic $3\times 10^{-3}\text{--}10^{-2}$ [5]	A tagging optics (β^* raised $\approx 15\times$ at $5 \times 41 \text{ GeV}$) with pots that follow, or IR-8 [37]; Z-identification against $\alpha+d$, both at beam rigidity; B0 calorimeter veto of the 3.56 MeV de-excitation γ

Table 3 — beam requirements and one-year reach by observable, from the programme's fast simulation and event-generator pseudo-experiments [5]. The Δ and coherent rows are at $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV}/u)$; the other observables combine the three beam configurations. All structure-function inputs are scenarios with explicit bands.

Species by species: ${}^6\text{Li}$ is the only $A > 2$ nucleus with $J \geq 1$ the EIC could polarize — b_1 and Δ do not exist for protons or ${}^3\text{He}$ at all — and the lattice indication that the deuteron's exotic glue is suppressed [49] makes a bound nucleus the likelier place for a discovery [3]. ${}^7\text{Li}$ is the only bound polarized proton within reach of any collider, and the only light-ion beam whose spectator is a spin-0, tightly bound nucleus landing in the Roman-Pot window today. Figures 2 and 3 show the phase space and the projected Δ measurement that Table 3 summarizes.

(x, Q^2) phase space and analysis binning — $e(10) \times {}^6\text{Li}(99.5/u)$, 1-year EIC program ($10 \text{ fb}^{-1}/u$), 40×30 log-log grid
color: expected events per bin where ≥ 1 ; e' cuts $|\eta| \leq 3.5$, $E' \geq 0.5 \text{ GeV}$, $0.01 \leq y \leq 0.95$, $W^2 \geq 10 \text{ GeV}^2$

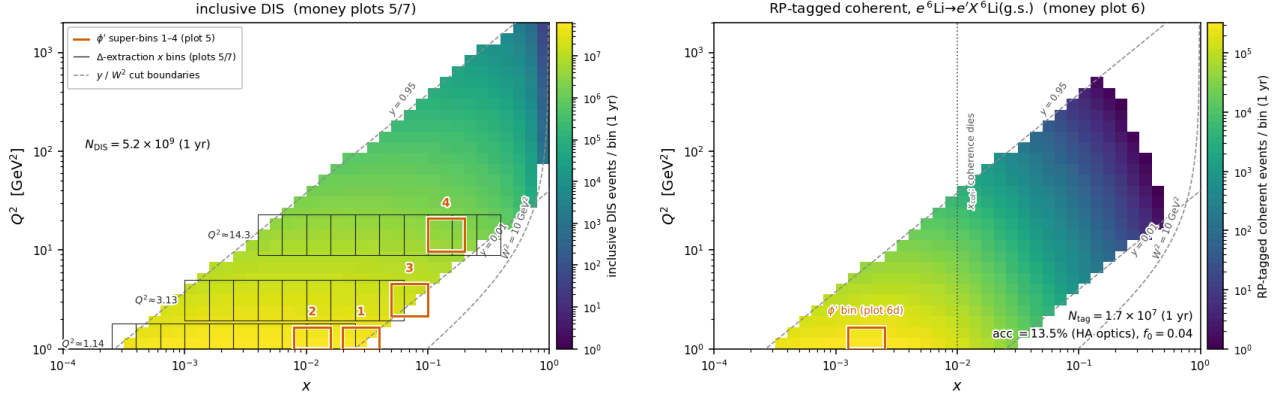


Figure 2 — expected events per (x, Q^2) bin in one EIC year at $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV}/u)$. Left: inclusive DIS (5.2×10^9 accepted events) with the bins of the $\cos 2\phi$ analysis overlaid; the measurement is limited by polarization, acceptance and systematics, not by statistics. Right: the coherent intact- ${}^6\text{Li}$ channel for a tag with a 0.20 GeV near-beam envelope (1.7×10^7 tagged events), confined to $x \leq 0.03$ because coherence requires the photon to resolve the whole nucleus, which fails once $x \geq 0.01$.

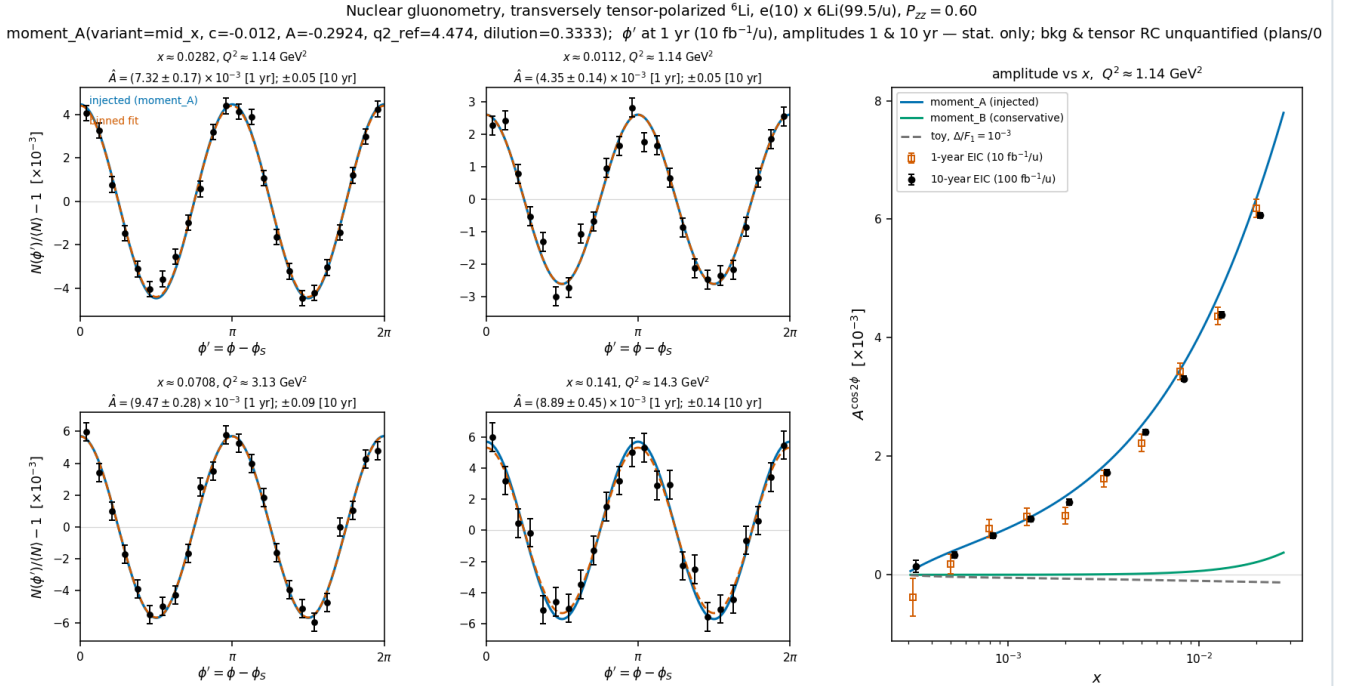


Figure 3 — projected extraction of $\Delta \cdot \cos 2\phi'$ pseudo-data (one EIC year, $P_{zz} = 0.6$) in the four highest-significance (x, Q^2) bins, with the injected moment-constrained Δ model and the binned fit; right, the extracted amplitude versus x against alternative model readings. The precision distinguishes between the models rather than only from zero. The reconstructed-level version uses the spin-state ratio in reconstructed bins [5].

7 • Summary

Lithium is the practical polarized nucleus for the EIC because three things coincide: an alkali atom that a single laser polarizes and an existing EBIS charge-breeds; two stable isotopes whose nuclear spin sits on a light cluster — a polarized deuteron in ${}^6\text{Li}$, a polarized proton in ${}^7\text{Li}$ — bound to an inert α that exact few-body theory describes; and ions whose spin dynamics in the ring are deuteron-like. Its physics is inaccessible without polarization, since b_1 , Δ , g_1^A and the tagged tensor asymmetries all vanish or are undefined without it, and largely inaccessible at a fixed target, where dilution, spin-temperature-limited tensor polarization and the loss of slow spectators are structural. Four measurements follow, each a first: b_1 of a nucleus beyond the deuteron, the never-measured double-helicity-flip structure function Δ , the polarized EMC

effect on a bound proton at collider kinematics, and α -tagged DIS on a polarized cluster. The requirements they place on the machine are specific — tensor polarization delivered as $m = \pm 1$ -rich and $m = 0$ -rich bunches alternating bunch by bunch, transverse polarization at the interaction point, relative luminosity between fills controlled to 10^{-4} , and a near-beam tagging capability for $A/Z = 2$ recoils that no published optics provides — and each is a concrete question to the accelerator and detector groups. The projections in the companion report [5] quantify what answering them buys.

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Appendix A · Revision history

DATE	REVISION
2026-08-17	First version: the DIS and spin-1 formalism from first principles, and the four flagship measurements (b_1 , Δ , the polarized EMC effect, spectator-tagged DIS) with their reach.
2026-08-21	§1 rewritten against primary sources — why lithium as a source and as a target, why polarization is indispensable, fixed target versus collider — and the requirements table added.
2026-08-25	References verified against the local copies in refs/ (two citations corrected); the spin-state ratio and bunch-by-bunch alternation as the Δ requirement; the two-state error formula.
2026-08-26	Development run 8 folded in: the PYTHIA 8 hadronic final state, the Roman-Pot aperture measured in the ePIC geometry, R from the R1998 fit, physical nuclear masses in the far-forward routing.
2026-08-27	Rewritten as a paper: abstract, a single logical thread (atom $\rightarrow$ nucleus $\rightarrow$ ion; polarization $\rightarrow$ collider; formalism $\rightarrow$ observables $\rightarrow$ requirements), references numbered in citation order, and every repeated fact stated once. All numbers restated from the current scripts at the γ -matched beam energies (^6Li at 40.8 / 99.5 / 137.5 GeV/u, plans/10): the sweet spots, amplitudes and errors of the Δ projection, the R-model shifts, the b_1 and polarized-EMC reach, and the coherent-channel acceptance, which with the Yellow Report divergences does not survive at IP6 with any published optics. The pre-2026-08-27 tagging and coherent numbers, computed at the superseded rigidity-scaled energies (13.5% at 50 GeV/u; the aperture results at 20.5 GeV/u), are withdrawn.

Companion documents: Report 1 (the projected $\cos 2\phi$ measurements, inclusive and coherent), Report 2 (the reconstruction chain and the reconstructed-level results), Report 3 (the machine and the detector). Figures from `evgen/scripts/phase_space_bins.py` and `evgen/scripts/money_cos2phi.py`; reach numbers from `fastsim/scripts/ ( tagging_acceptance.py , money_b1.py , money_polemc.py , money_delta.py )` and the reconstructed-level scripts of `evgen/scripts/`; every command is in `docs/reproduction_manual.md`. Generated by `reports/build_report.py`; display mathematics typeset at build time (matplotlib mathtext, STIX). Projections use scenario structure functions with explicit bands.